Topological Analysis of pLDDT Profiles: Reproducing and Extending Cazals & Sarti (2025)

Algorithms in Structural Bioinformatics — Q4 Project

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Outline

1. Motivation
2. Mathematical Framework
3. Pipeline and Algorithm
4. Arity Extension (Q1 $\times$ Q4)
5. Validation
6. Proteome-Scale Results
7. Discussion
8. Conclusions

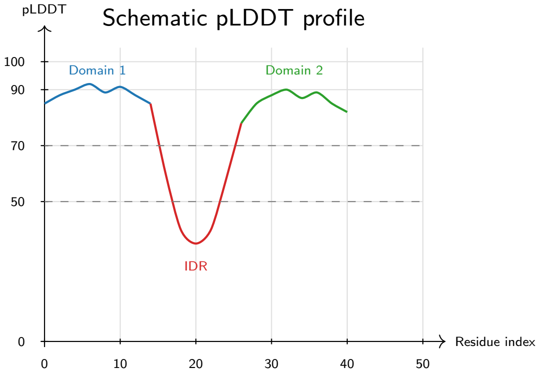
What is pLDDT and why does its *structure* matter?

pLDDT (predicted Local Distance Difference Test)

- Per-residue confidence score, $s_i \in [0, 100]$
- Stored in the B-factor column of AlphaFold PDB files
- $s_i \gtrsim 70$: well-folded region
- $s_i \lesssim 50$: intrinsically disordered

Key observation

- Most analyses use pLDDT as a *threshold* (keep/discard)
- **Cazals & Sarti (2025)** treat the full pLDDT *profile along the chain* as a topological object
- This encodes multi-domain and disordered-linker structure



Project goal

Reproduce the full Q4 fragmentation pipeline on the complete *H. sapiens* proteome (23,391

Path graph and scalar filtration

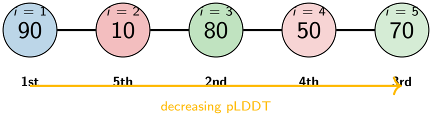
Model: protein chain $\to$ **path graph** $G_n = (V, E)$

- Vertices $V = \{1, \dots, n\}$: residues
- Edges $E = \{(i, i{+}1)\}$: peptide bonds
- Scalar function: $u_i = -s_i$ (negative pLDDT)

Filtration: insert vertices in increasing u order (= **decreasing pLDDT** order)

$\emptyset \subseteq \mathcal{G}_{u_1} \subseteq \cdots \subseteq \mathcal{G}_{u_n} = G_n$

An edge $(i, i{+}1)$ is **activated** the moment *both* endpoints are inserted.



Numbers = pLDDT. Insertion order = by decreasing pLDDT.

Connected components, Elder Rule, and Persistence Diagram

$N_{cc}(u)$: number of connected components at level u

- New vertex inserted $\Rightarrow N_{cc} + 1$
- Edge joins two components $\Rightarrow N_{cc} - 1$

Elder Rule: when two components merge, the one born at *higher* pLDDT (older) survives. The younger one **dies**, producing a pair:

$$(b_i,\ d_i) \quad b_i \geq d_i \geq 0$$

Persistence: $p_i = b_i - d_i$
Accretion: $p_i = 0$ (same-level merge)

Persistence Diagram (PD)

Collection of all $m = n - 1$ pairs $\mathcal{P} = \{(b_i, d_i)\}_{i=1}^{n-1}$

Toy example: pLDDT $= [90, 10, 80, 50, 70]$

Event	N_{cc}	PD pair
Insert $i = 1$ (90)	1	—
Insert $i = 3$ (80)	2	—
Insert $i = 5$ (70)	3	—
Insert $i = 4$ (50), edge $(4, 5)$	3	$(70, 50)$
edge $(3, 4)$	2	$(80, 50)$
Insert $i = 2$ (10), edge $(1, 2)$	2	$(10, 10)^\dagger$
edge $(2, 3)$	1	$(10, 10)^\dagger$

†Accretions — zero persistence

Positive persistence pairs: $(70, 50)$ and $(80, 50)$

Five topological statistics

Let $m = n-1$, $m' =$ positive-persistence pairs, $\mathcal{P}_D =$ set of distinct persistence values.

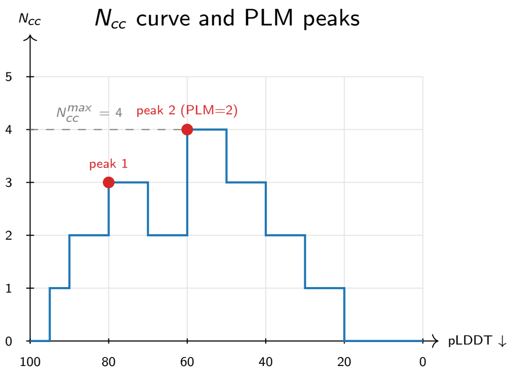
1. $f_{cp}^{+} = \dfrac{m'}{n-1}$
   Fraction of heterogeneous transitions

2. $\bar{p} = \dfrac{1}{m}\sum_i (b_i - d_i)$
   Mean persistence lifetime

3. $H_p = \dfrac{-\sum_v \mathbb{P}[v] \ln \mathbb{P}[v]}{\ln |\mathcal{P}_D|}$
   Normalised Shannon entropy of persistence
   $H_p \in [0,1]$; high = diverse pLDDT profile

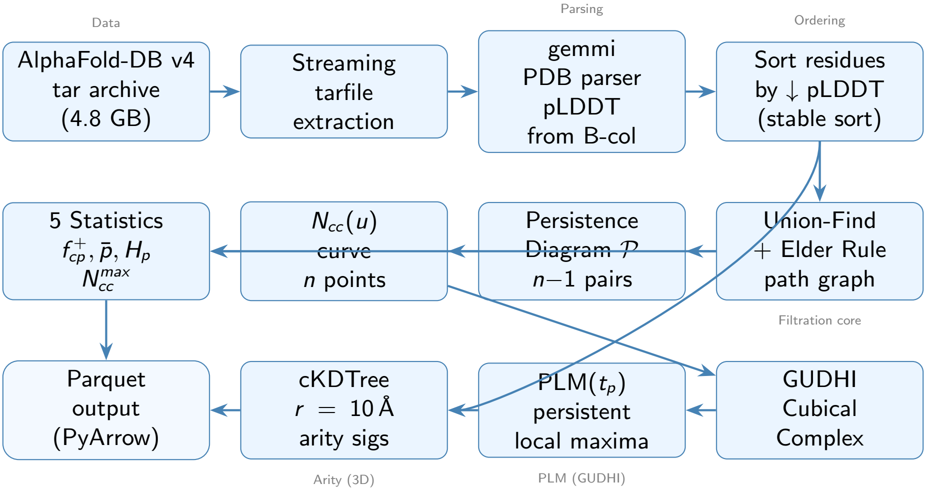
4. $N_{cc}^{max} = \max_u N_{cc}(u)$
   Peak simultaneous fragmentation

5. $\mathrm{PLM}(t_p)$: persistent local maxima of N_{cc}
   Number of significant domain peaks



PLM ≥ 3 at $t_p = 0.025$ flags a protein as a **multi-domain fragmentation candidate**.

End-to-end pipeline overview



6 parallel workers via `multiprocessing.Pool` — processes 23 391 proteins in ≈30 seconds.

Algorithm: standard sublevel-set filtration

Path-graph pLDDT filtration (paper-faithful)

Input: $(s_1, \ldots, s_n)$, threshold t_p **Output:** $\mathcal{P}$, N_{cc}-curve, $\mathrm{PLM}(t_p)$

1. $\pi \leftarrow \textsc{StableArgsort}(-s)$ *decreasing pLDDT; ties by index*
2. Init UF: $\mathrm{parent}[i] \leftarrow i$, $\mathrm{birth}[i] \leftarrow -\infty$, $\mathrm{inserted}[i] \leftarrow \bot$; $N_{cc} \leftarrow 0$
3. **for each** i in π: $\mathrm{birth}[i] \leftarrow s_i$; $\mathrm{inserted}[i] \leftarrow \top$; $N_{cc} \leftarrow N_{cc} + 1$
   - **for** $j \in \{i-1, i+1\}$ with $\mathrm{inserted}[j]$; let $r_i, r_j = \textsc{Find}(i, j)$:
     - $r_i = r_j$: append (s_i, s_i) (accretion)
     - else (**Elder Rule**): survivor $= \arg\max \mathrm{birth}$; merge loser$\rightarrow$survivor; $N_{cc} -= 1$; append $(\mathrm{birth}[\text{loser}], s_i)$
   - append (s_i, N_{cc}) to Curve
4. $\mathrm{PLM}(t_p) \leftarrow \textsc{GUDHIPersistentMaxima}(\text{Curve}, t_\nu = n \cdot t_p)$
5. **return** $\mathcal{P}$, Curve, $\mathrm{PLM}(t_p)$

$\textsc{Find}$ uses path-halving $(O(\alpha(n)) \approx O(1))$. GUDHI step: negate N_{cc}, 1D cubical complex, count 0-dim pairs with persistence $\geq t_\nu$.

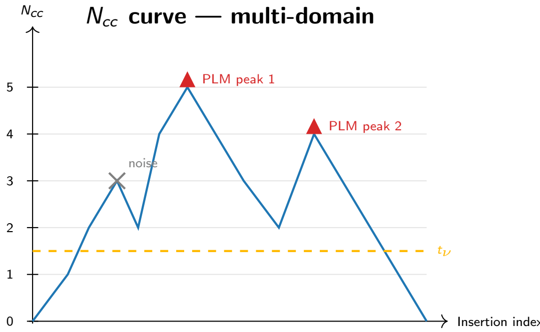
PLM via GUDHI — second-layer persistence

The N_{cc} curve is itself a 1D scalar function.
Its local maxima correspond to **pLDDT domain peaks**.

How we count significant peaks:

1. Negate N_{cc}: local maxima become local *minima*
2. Feed $-N_{cc}$ to GUDHI's `CubicalComplex`
3. Each 0-dim pair (b_j, d_j) in the resulting PD has persistence $= N_{cc}[\text{peak}] - N_{cc}[\text{valley}]$
4. Count pairs with persistence $\geq t_\nu = n \cdot t_p$

Why two layers of persistence?
First layer: filtration of the *protein graph*.
Second layer: Morse-Smale simplification of the N_{cc} *function* — filters out noise peaks from float-precision pLDDT.



Arity: a 3D packing descriptor

Arity of residue k:
$\#$ of C_α atoms within 10 Å of $C_\alpha(k)$ (excluding self)

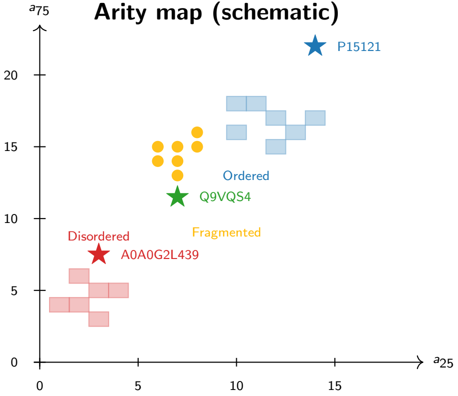
Arity signature (a_{25}, a_{75}):
25th and 75th percentiles of per-residue arity

- a_{25} low $\Rightarrow$ flexible/disordered regions dominate
- a_{75} high $\Rightarrow$ well-packed globular regions present
- Multi-domain proteins: *both* — compact domains + disordered linkers

Implementation:

- Extract C_α coordinates (gemmi)
- Build `scipy.cKDTree`
- `query_ball_point(r=10)` — $O(n \log n)$
- Subtract 1 for self-count

Arity map (schematic)



Validation: toy example and null model

Toy protein ($n = 5$, pLDDT $= [10, 90, 80, 90, 10]$)

Pair	(b_i, d_i)	Type
$(80, 80)$	accretion	
$(90, 80)$	$p = 10$	positive
$(10, 10)$	accretion	
$(10, 10)$	accretion	

Hand-computed:

$$H_p = 0.8112, \quad f_{cp}^{+} = 0.25$$

Code output (4 d.p.):

$$H_p = 0.8113, \quad f_{cp}^{+} = 0.25 \checkmark$$

Null random model ($n = 1{,}000$, pLDDT $\sim \mathcal{U}[0, 100]$)

Cazals & Sarti (2025) Example 1:

- $f_{cp}^{+} \approx 1/3 = 0.333$
- $H_p \approx 0.45$
- $N_{cc}^{max} \approx n/4 = 250$

Our result (raw float pLDDT):

- $f_{cp}^{+} = 0.329 \checkmark$
- $H_p = 0.438 \checkmark$
- $N_{cc}^{max} = 252 \checkmark$

Note: despite continuous float inputs, only 330 distinct persistence values arise, keeping $H_p \ll 1$.

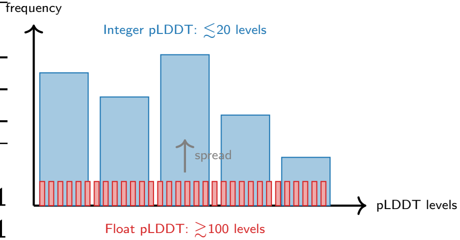
Validation: three prototype proteins

UniProt	Type	n	H_p	f_{cp}^{+}	PLM
Paper targets (Fig. 3)					
P15121	Ordered	316	$\approx$0.04	—	—
A0A0G2L439	Disord.	449	$\approx$0.27	—	—
Q9VQS4	Mixed	781	$\approx$0.28	—	—
Our results (raw float pLDDT)					
P15121	Ordered	316	0.371	0.241	1
A0A0G2L439	Disord.	449	0.466	0.339	1
Q9VQS4	Mixed	781	0.424	0.312	2

Key: relative ordering preserved

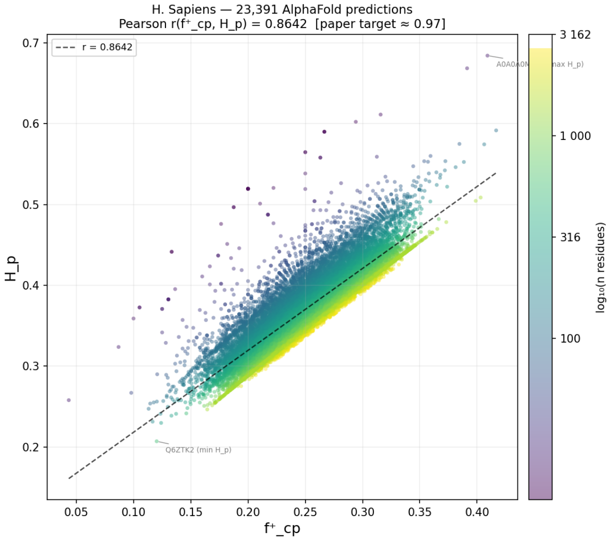
Ordered $<$ Mixed $<$ Disordered ✓

Why are our H_p values higher?



More distinct pLDDT levels $\rightarrow$ more distinct persistence values $\rightarrow$ broader $\mathbb{P}[v]$ $\rightarrow$ higher H_p.

Proteome-scale Pearson correlation (Figure 7)



Setup: all 23 391 AlphaFold-DB v4 fragments

No length pre-filter (paper-faithful)

Metric	Value
N	23 391
Pearson r	**0.864**
Paper target	≈ 0.97
Gap	-0.106

Both statistics elevated (diagonal shift)

$\rightarrow$ slope preserved, variance increased

$\rightarrow$ r reduced by float-precision effect

Median $f^{+}_{cp} = 0.242$, Median $H_p = 0.362$

Fragmentation candidates and t_p sensitivity

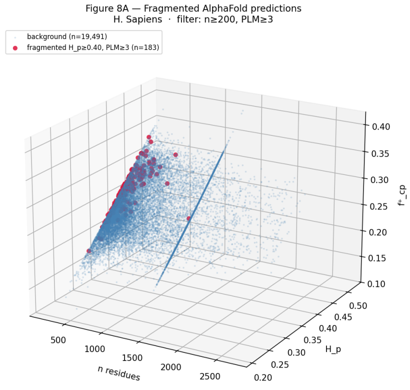
Three-way filter (paper's Fig. 8):

$$n \geq 200, \quad H_p \geq 0.25, \quad \mathrm{PLM}(t_p) \geq 3$$

t_p	Candidates	Paper
0.020	822	—
0.025	**183**	**≈86**
0.030	34	—

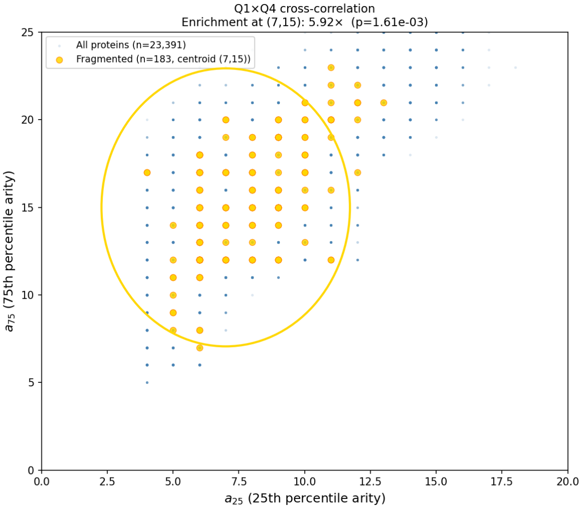
Why 183 instead of 86?
Float pLDDT creates many fine-grained N_{cc} oscillations → more peaks clear the threshold $t_\nu = 0.025 \cdot n$

34 proteins survive all thresholds
= **robust, threshold-independent core**



3D scatter: length × H_p × f^+_{cp}. Orange = fragmentation candidates.

Q1 × Q4 cross-correlation



Finding: fragmented proteins (PLM ≥ 3) cluster in a specific arity bin

Property	Value
Centroid	$(a_{25} = 7,\ a_{75} = 15)$
Enrichment	**5.92×**
p-value	1.61×10^{-3}
Test	Fisher exact

Biological interpretation

- $a_{25} \approx 7$: disordered linkers with few contacts
- $a_{75} \approx 15$: well-packed domain cores
- Intermediate arity = mix of compact domains + disordered linkers = **multi-domain hallmark**

This finding is *orthogonal* to pLDDT — derived purely

The float-precision gap — unified explanation

Root cause
AlphaFold-DB v4:
float pLDDT ($\gtrsim$100 levels)
vs. paper's $\lesssim$20 int levels

More distinct
persistence values
in $\mathcal{P}_D$

H_p inflated
($+0.14$ to $+0.33$)

Pearson r:
0.864 vs 0.97

Finer N_{cc} curve
more local
oscillations

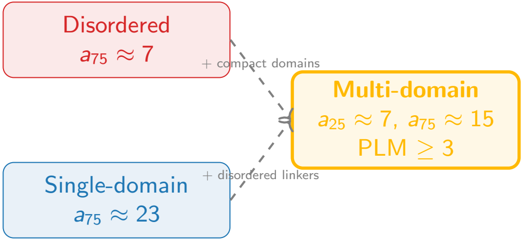
More PLM peaks
survive t_ν

Candidates:
183 vs 86

Both gaps share a single falsifiable hypothesis: reproducing the analysis on integer rounded pLDDT would close

Arity enrichment: biological significance

What the enrichment tells us



Two orthogonal signals agree:

1. **pLDDT fragmentation** (Q4):
   PLM ≥ 3 identifies proteins with multiple confident domains separated by linkers
2. C_α **arity** (Q1):
   Moderate a_{25} (linkers) + moderate a_{75} (domains) $\rightarrow$ same proteins

Convergence of two independent structural descriptors provides mutual validation and strengthens confidence in the topological fragmentation signal.

Summary

What we reproduced

- Full path-graph filtration from scratch (Union-Find + Elder Rule)
- All 5 statistics + PLM via GUDHI
- Applied to all 23 391 H. sapiens AlphaFold-DB v4 fragments
- Pearson $r = 0.864$ vs target 0.97
- 183 fragmentation candidates vs target 86
- Relative ordering of prototypes preserved ✓
- Null model targets matched ✓

Novel extension (Q1 × Q4)

- Proteome-wide arity signatures
- 5.92-fold enrichment of fragmented proteins at $(a_{25} = 7, a_{75} = 15)$
- $p = 1.61 \times 10^{-3}$ (Fisher exact)
- Orthogonal 3D validation of pLDDT fragmentation signal

Key insight

All discrepancies with the paper trace to a single cause: float vs. integer pLDDT in AlphaFold-DB v4 vs. the paper's data.

Code: Python 3.11 — NumPy · SciPy · gemmi · GUDHI · PyArrow · streaming tar pipeline (30 s for 23 391 proteins)

Thank you

Questions?

Paper: Cazals & Sarti (2025), bioRxiv 2024.11.16.623929
Data: AlphaFold-DB v4 — *H. sapiens* proteome
Code: Python, available with the submission